Karthik Seetharaman

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Education

Stanford University Stanford, CA
B.S. Mathematics, B.S. Computer Science (Systems), GPA: 4.07 *Sep. 2022 – Jun. 2026*
Relevant Coursework: *Machine Learning, Convex Optimization, Parallel Computing, Language Modeling*
Honors: *Honors in Computer Science (thesis on causal machine learning), Phi Beta Kappa, Tau Beta Pi*

Work Experience

Seed Startups Dec. 2025 - Present
AI Engineer *San Francisco, CA*
- **Kinetic Systems:** Creating realistic voice agents for healthcare support training and realtime voice-triggered database search using RAG, Vapi — Apr. 2026 - Present (part-time)
- **CompLabs:** Created multi-agent architecture for setting up and conducting laser-tracing simulations end-to-end, speeding up execution time by 20x — Dec. 2025 - Jan. 2026 (full-time)

Citadel Securities Jun. 2025 - Aug. 2025
Quantitative Research Intern *Miami, FL*
- Retail FX: analyzed data and created novel features from client-specific behavior to predict future flow, improving correlations with target by 20% over baseline non-client aware model
- Built out an efficient pipeline in Python (Pandas, Sklearn) to quickly test new statistical features and ML methods

Etched Oct. 2024 - Jun. 2025
Machine Learning Research & Host Software Intern *San Jose, CA*
- Created and implemented novel speculative decoding algorithms to speed up LLM inference in Python
- Achieved up to 11x speedup over autoregressive decoding on code generation (HumanEval dataset)
- Implemented a mock KV cache allocator with paged attention based on novel hardware architecture in Rust

Stanford AI Lab Feb. 2024 - Sep. 2024
Undergraduate Research Assistant *Stanford, CA*
- Developed novel algorithms for reinforcement learning with human feedback in a heterogeneous population, ran experiments in bandit and LLM settings using PyTorch and Huggingface to prove effectiveness
- **Second-author publication** at AISTATS 2026: *Direct Preference Optimization With Unobserved Preference Heterogeneity: The Necessity of Ternary Preferences*

Qualcomm Jun. 2023 - Sep. 2023
Embedded Software Engineer Intern *San Diego, CA*
- Automated DLL testing with Python and created a full-stack data pipeline, saving about 5 minutes per test

Research Experience & Publications

Asynchronous Stacking for LLM Pretraining Mar. 2026 – Present
Stanford University *Stanford, CA*
- Conducting research under Tatsu Hashimoto (**Stanford NLP Group**) on training LLMs with progressive layer stacking; utilizing PyTorch to experiment with pretraining methods and run jobs on Stanford GPU clusters

Patterns in the Lattice Homology of Seifert Homology Spheres Jan. 2021 – Jan. 2022
Massachusetts Institute of Technology *Cambridge, MA*
- **Won** Honorable Mention in the 2022 S.T. Yau Mathematics Award, top 10 out of over 1,000 projects submitted

The Rank-Generating Functions of Upho Posets Jan. 2020 – Jan. 2021
Massachusetts Institute of Technology *Cambridge, MA*
- **Published** in *Discrete Mathematics, Vol. 345, Issue 1*, marked Editors' Choice (top 15 articles in the last 3 years)

Honors & Awards

- **Top 200 2022/2023, Top 500 2024/2025 Putnam Competition** – Scored above 30/120 all four years and 41/120 on the 2023 Putnam exam, median score was 1/120
- **MIT PRIMES Scholar** – Top 30 high schoolers in the USA, selected to perform graduate-level math research

Skills

Programming: Python (Pandas, Sklearn, NumPy, PyTorch), C++, C, Java, SQL, LaTeX, Unix, Excel